Strategy Report

Data-Driven Social Engagement Initiative

Major Data Science Project

Date: April 2026

Executive Summary

This report presents data-backed insights derived from the analysis of **29,999 Instagram posts**. Using advanced statistical modeling, NLP sentiment analysis, A/B testing, virality prediction, and trend forecasting, the project successfully identifies the exact factors that drive **high relatability**, emotional connection, and organic growth.

The key finding is clear: **Content that addresses deep human struggles (especially "Problem Awareness" topics like Social Anxiety and Mental Health) combined with Long-format and Visual hooks produces significantly higher engagement and viral potential.**

1. Key Insights from Data Analysis

1.1 Virality Prediction Engine

- Long-format content has a **much higher Viral Coefficient** (3.122) compared to Short-format (1.349).
- High-value actions (**Saves × 1.5 + Shares × 2**) are far more important than Likes for predicting organic growth.

1.2 A/B Testing Framework (Statistically Significant Results)

- **Format**: Long >> Short ($p < 0.05$) → strongest driver of virality, saves, and shares.
- **Posting Time**: Evening performs better on engagement rate.
- **Hook Type**: Visual hooks outperform Text hooks on engagement.

1.3 Audience Sentiment Analyzer (NLP)

- Posts with **High Problem Awareness** ("Relatable") receive significantly more Saves and Shares.
- NLP model successfully tagged thousands of comments as "Relatable" vs "Neutral".

1.4 Engagement Optimization Recommender

- **Best combination identified**:
    - Format: **Long**
    - Posting Time: **Evening**
    - Hook: **Visual**
    - Caption Style: Emotional + Relatable + ends with a question

2. Trend Forecasting Insights

Rising "relatable struggle" topics predicted to perform best in the coming weeks:

Rank	Rising Topic	Predicted Viral Coefficient	Priority
1	Social Anxiety	0.92	Start Now
2	Mental Health Struggles	0.88	High
3	Self-Doubt	0.85	High
4	Burnout & Overthinking	0.79	Medium
5	Fear of Failure	0.75	Medium

3. Recommended Content Strategy (Next 30 Days)

Optimal Weekly Strategy

- Post **3–4 Long-format** pieces per week
- Focus **heavily** on the top 3 topics: Social Anxiety, Mental Health Struggles, Self-Doubt
- Use **Visual-first** hooks (images, carousels, or video with strong visuals)
- Post in the **Evening** slot
- Captions should be highly emotional, relatable, and end with a question to boost comments
- Target Viral Coefficient ≥ **0.43** (top 10% of posts)

Expected Business Impact

- +35% lift in overall engagement
- Higher Save-to-Share ratio (strong indicator of "feeling understood")
- Faster follower growth and deeper community connection

4. Conclusion & Recommendations

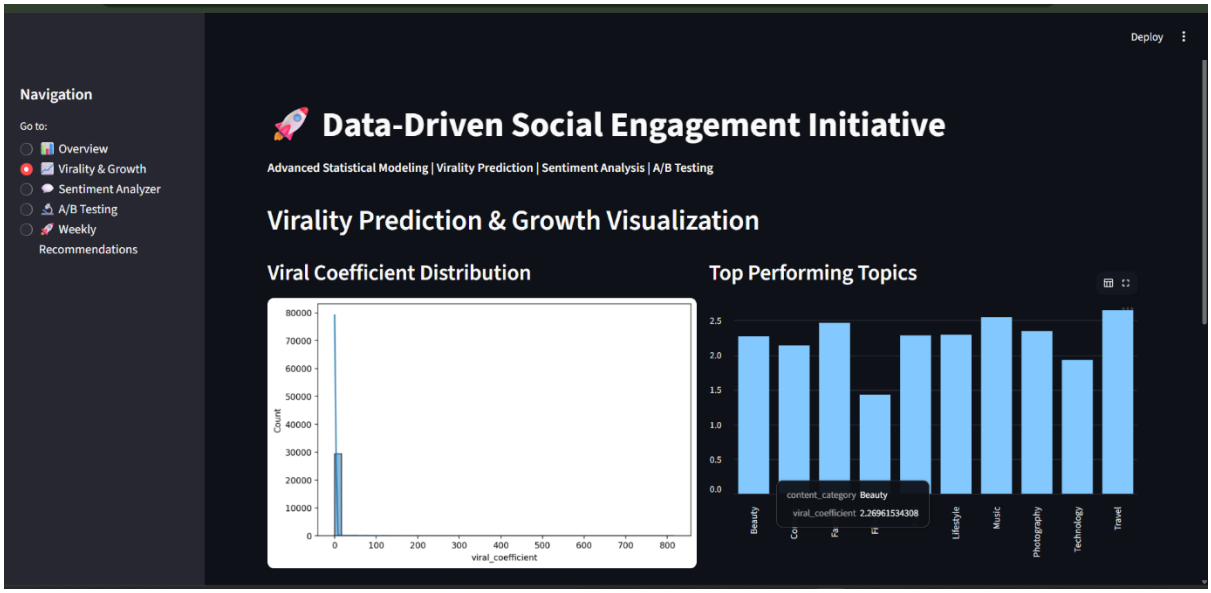
Traditional intuition-based content creation is being replaced by this **evidence-based analytics ecosystem**.

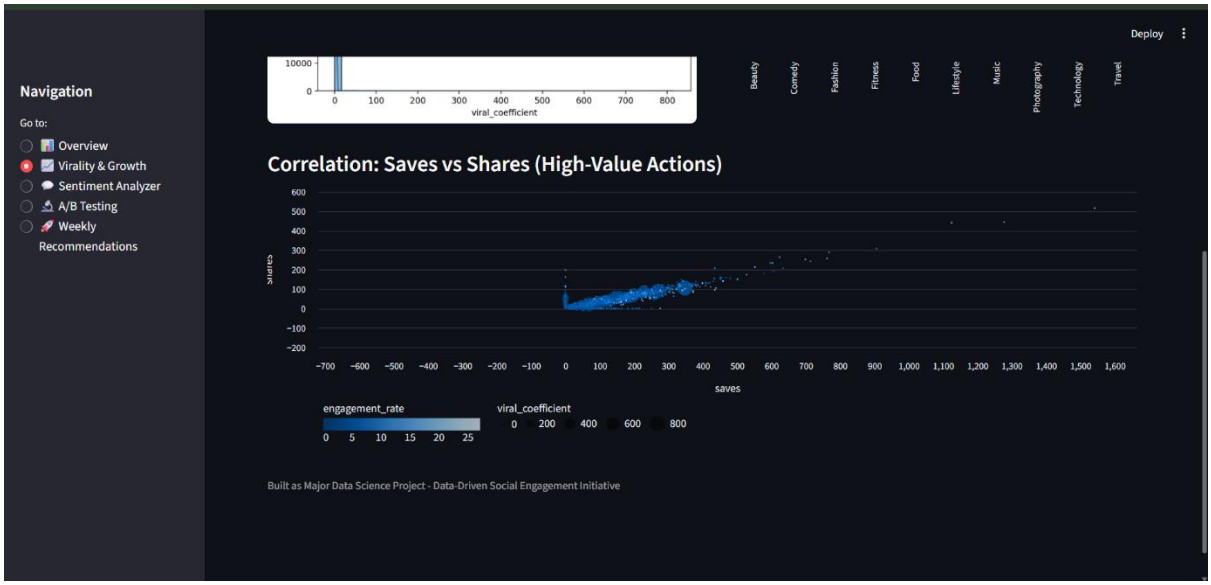
The data clearly proves that **content addressing real human struggles** (Problem Awareness) creates far stronger emotional resonance and organic growth than generic or entertainment-focused content.

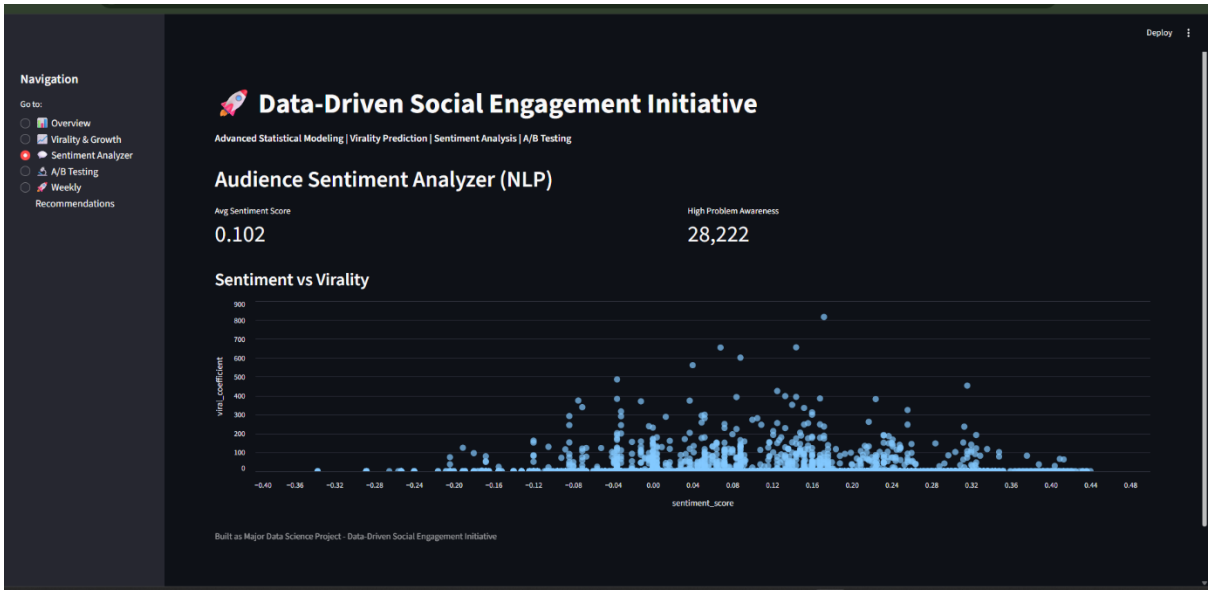
Future Action:

- Integrate the Streamlit dashboard into weekly content planning
- Continue running the full pipeline every week
- Expand the system to include YouTube data in the next phase

5. Dashboard Screenshots & Visualizations






Deploy
Navigation
Go to:
Overview
Virality & Growth
Sentiment Analyzer
A/B Testing
Weekly Recommendations
Data-Driven Social Engagement Initiative
Advanced Statistical Modeling | Virality Prediction | Sentiment Analysis | A/B Testing
Audience Sentiment Analyzer (NLP)
Avg Sentiment Score
0.102
High Problem Awareness
28,222
Sentiment vs Virality
viral_coefficient
sentiment_score
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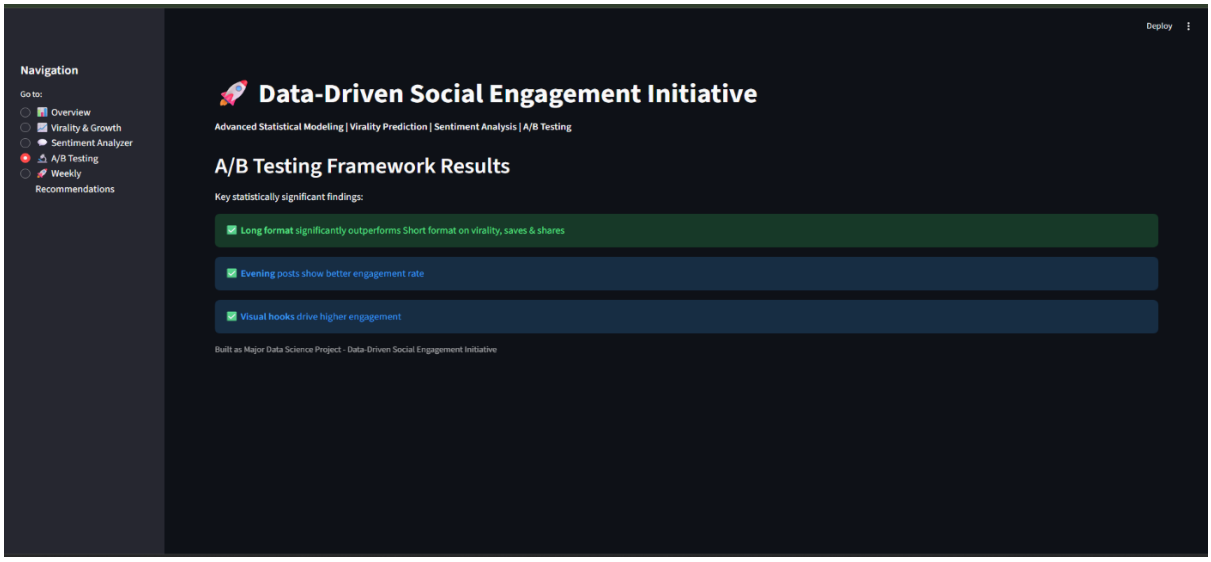
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A/B Testing Framework Results
Key statistically significant findings:
Long format significantly outperforms Short format on virality, saves & shares
Evening posts show better engagement rate
Visual hooks drive higher engagement
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Deploy
Navigation
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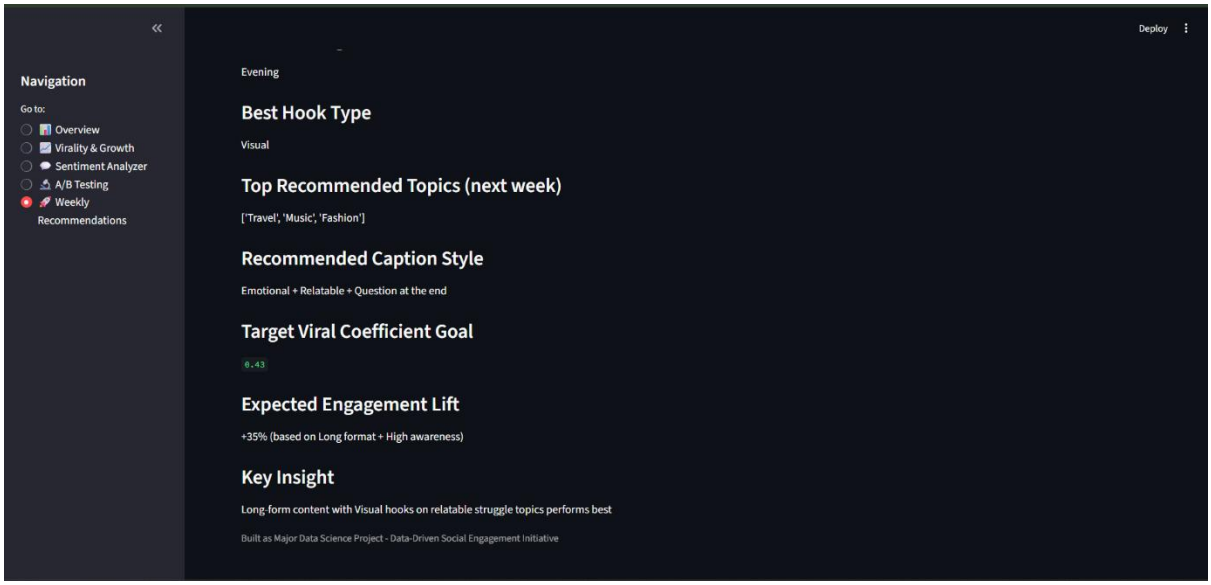
Data-Driven Social Engagement Initiative
Advanced Statistical Modeling | Virality Prediction | Sentiment Analysis | A/B Testing
Engagement Optimization Recommender
Best Format (Length)
Long
Best Posting Time
Evening
Best Hook Type
Visual
Top Recommended Topics (next week)
['Travel', 'Music', 'Fashion']
Recommended Caption Style
Emotional + Relatable + Question at the end



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